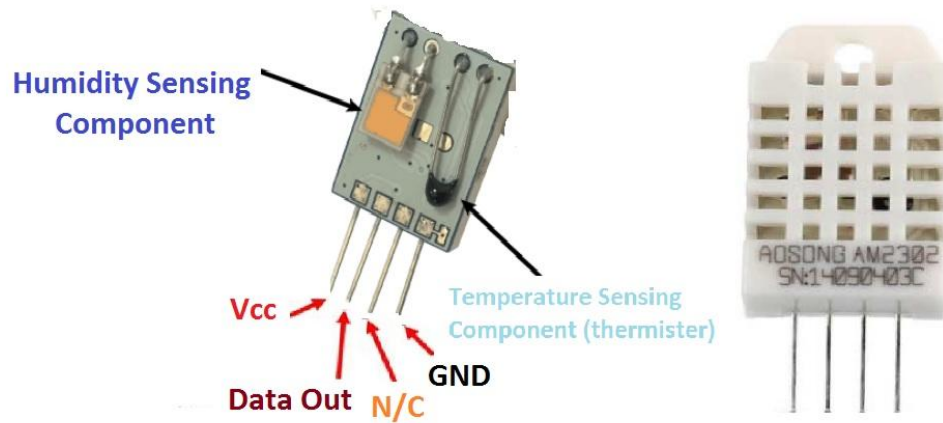
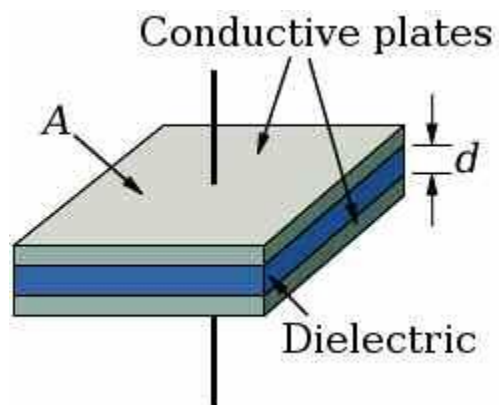


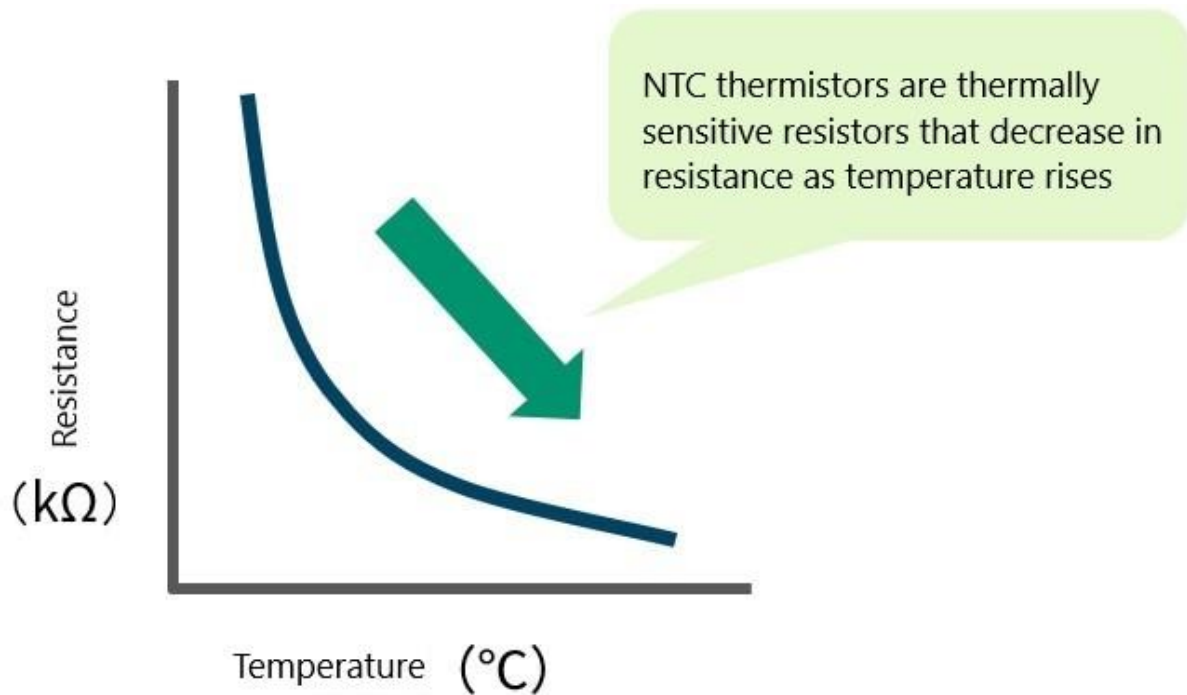
1 DHT22 – Temperature & Humidity Sensor

Introduction to DHT22



www.TheEngineeringProjects.com





4

Scientific Principles:

✓ (A) Humidity → Capacitive Sensing

- Inside DHT22 is a **polymer dielectric material** placed between two electrodes.
- When humidity increases:
 - Water molecules are absorbed in the polymer.
 - **Dielectric constant changes.**
 - Capacitance changes.

Formula:

$$C = \frac{\epsilon A}{d}$$

Where:

- ϵ = dielectric constant
- A = area
- d = distance

Since humidity changes ϵ , capacitance changes.

✓ (B) Temperature → NTC Thermistor Principle

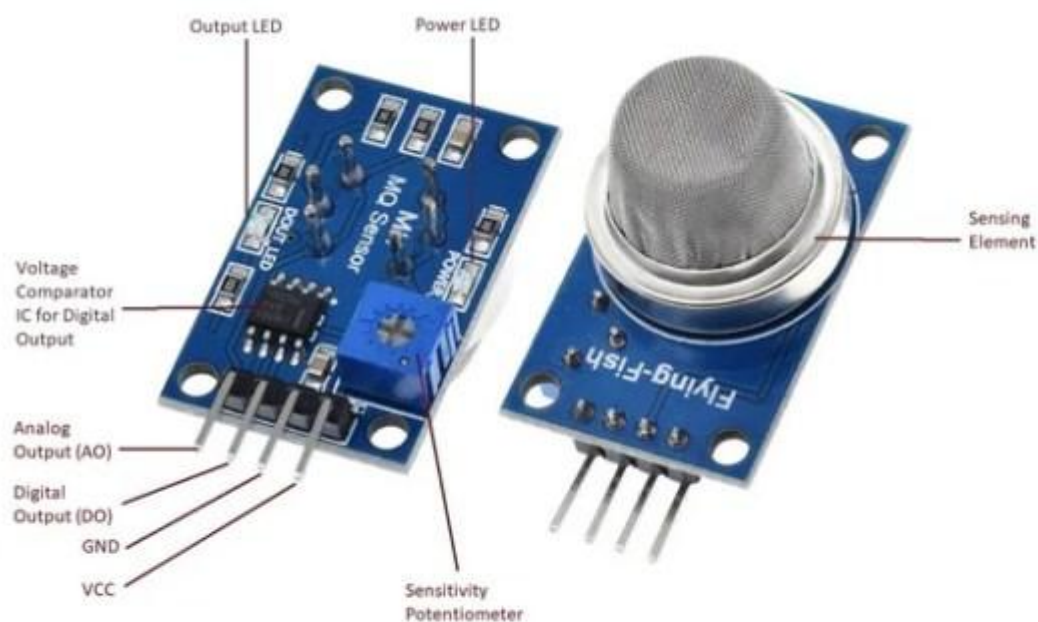
- Uses **Negative Temperature Coefficient (NTC) thermistor**.
- As temperature increases → resistance decreases.

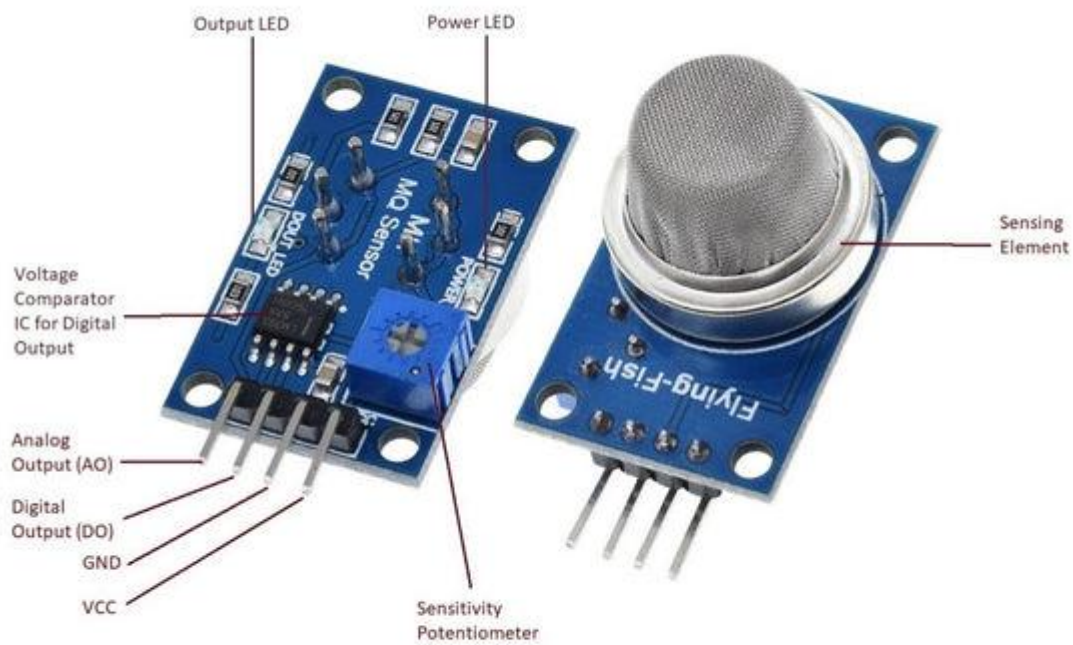
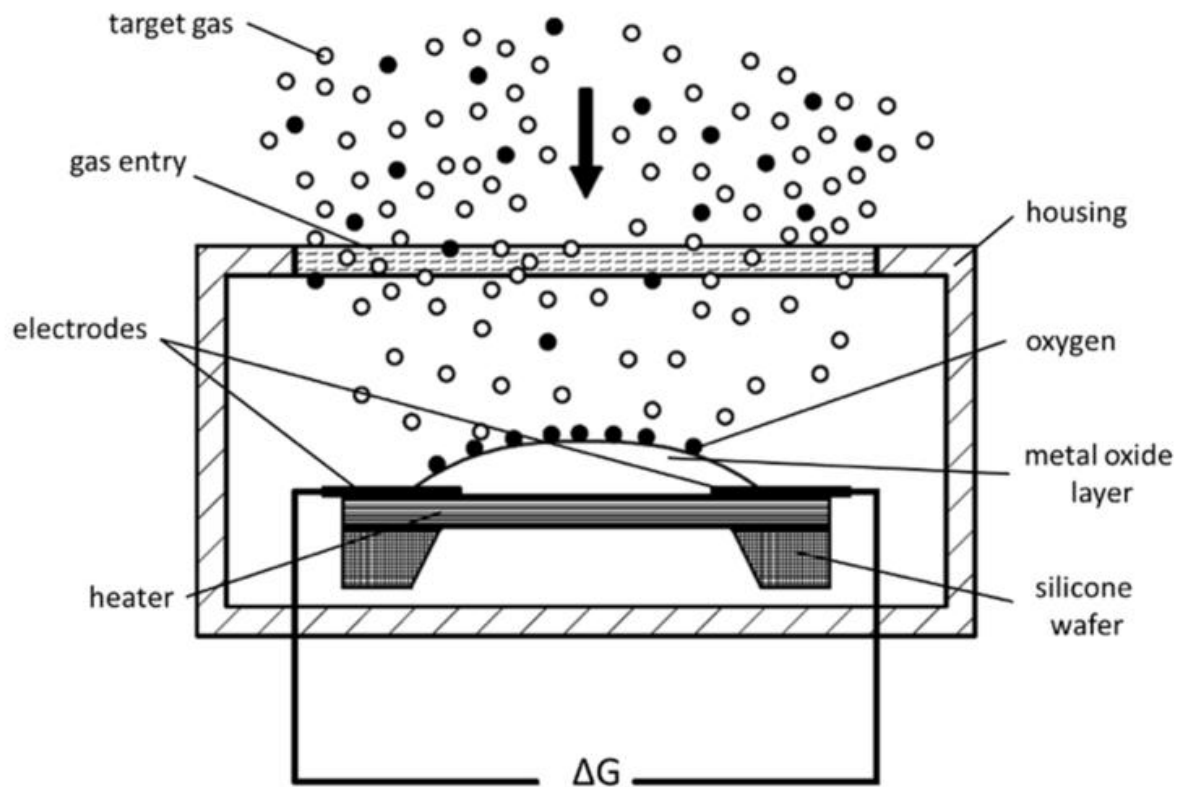
✚ Based on:

$$R_T = R_0 e^{B\left(\frac{1}{T} - \frac{1}{T_0}\right)}$$

This resistance change is converted to digital output.

🔥 2 MQ-2 Gas Sensor





Scientific Principle:

✓ Metal Oxide Semiconductor (MOS) Gas Sensing

- Sensing material: **Tin dioxide (SnO_2)**.
- Heated by internal heater coil ($\sim 300^\circ\text{C}$).

How it works:

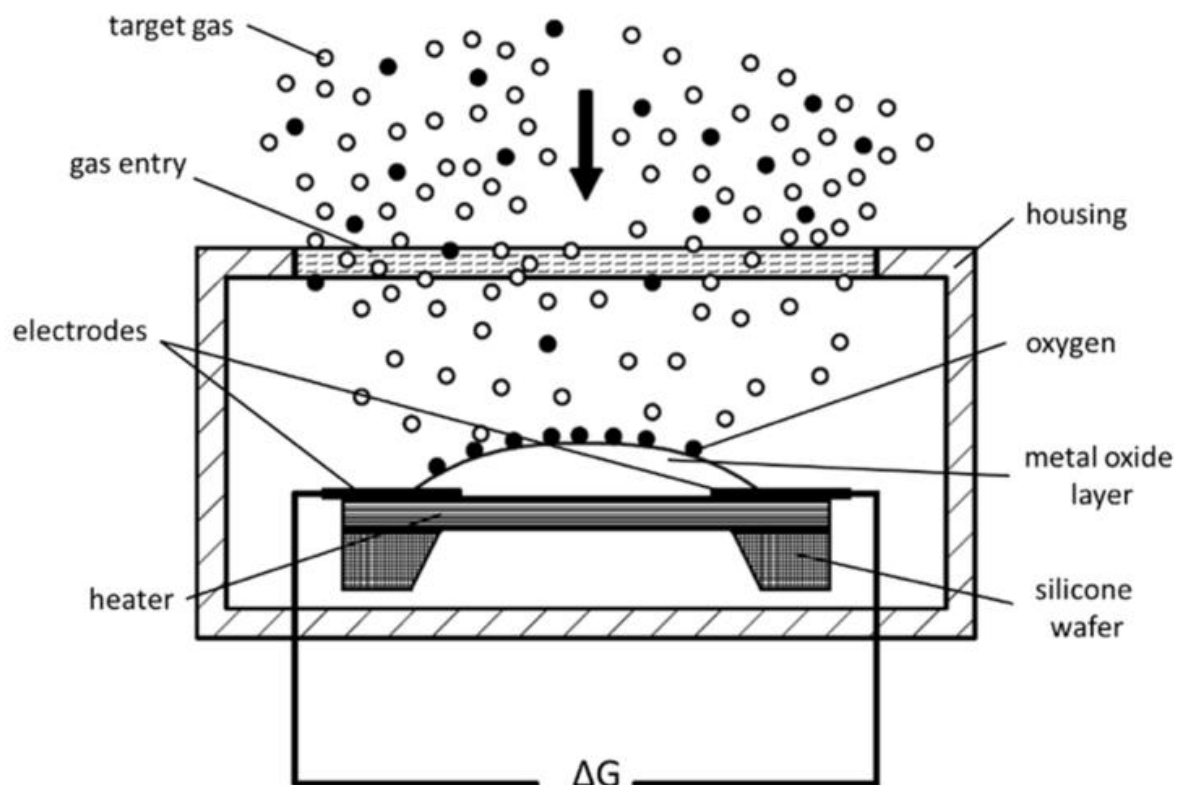
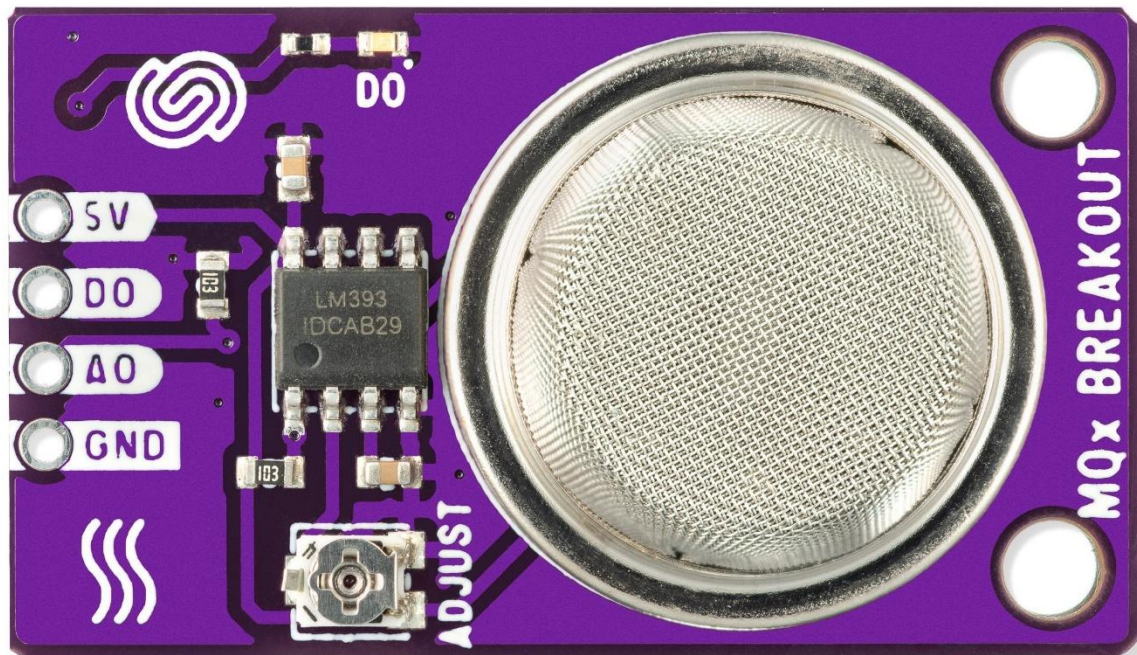
1. In clean air:
 - Oxygen molecules adsorb on SnO_2 surface.
 - They capture electrons $\rightarrow$ resistance increases.
2. When gas (LPG/smoke) present:
 - Gas reacts with adsorbed oxygen.
 - Electrons released back.
 - Resistance decreases.

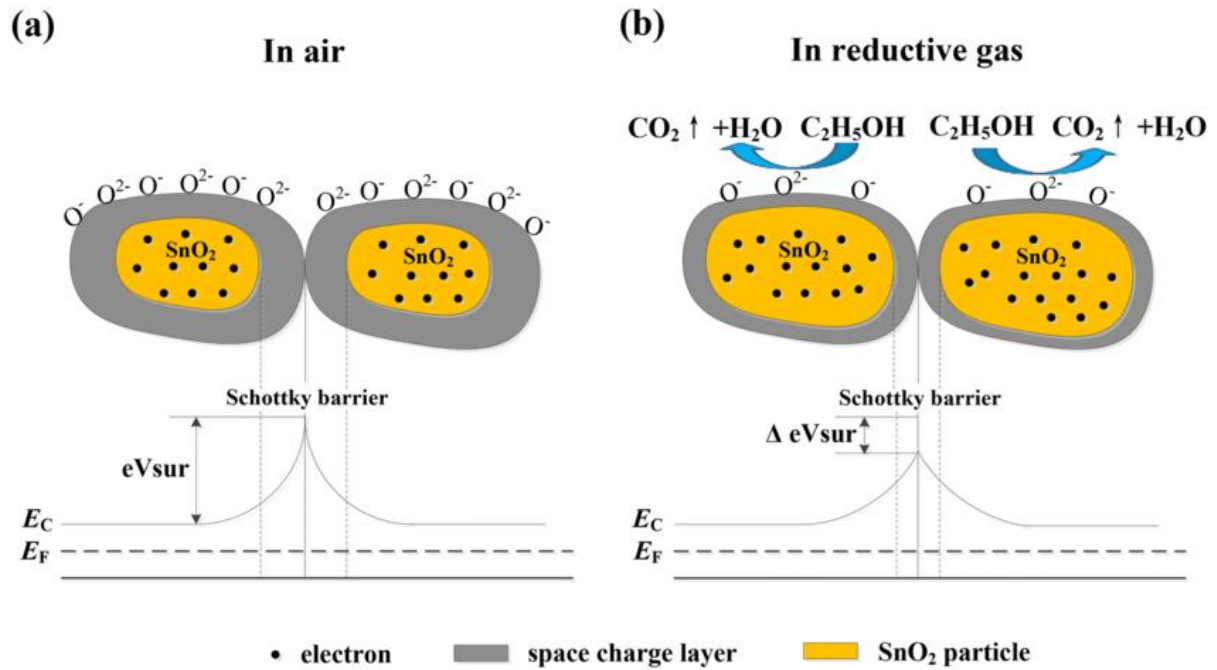
 So:

Gas concentration $\propto$ Change in resistance

This is called **Surface Chemical Reaction + Conductivity Change**.

3 MQ-135 Air Quality Sensor





4

Scientific Principle:

Same as MQ-2:

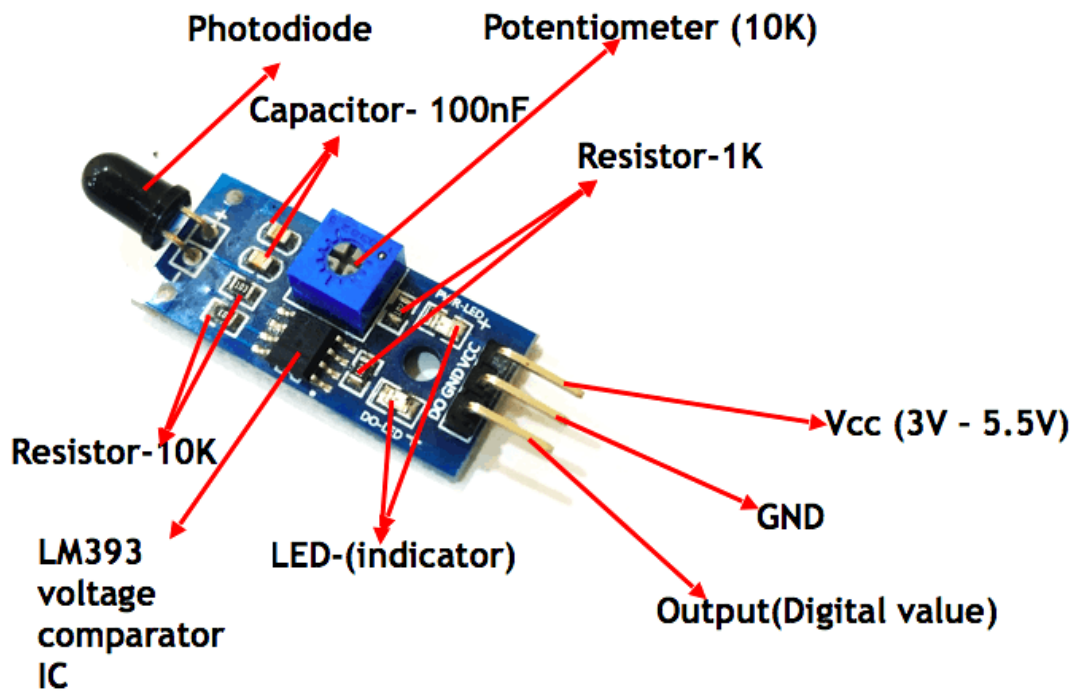
Semiconductor Gas Sensing via Chemisorption

- Uses heated SnO₂ material.
- Pollutant gases react with oxygen layer.
- Causes change in electrical conductivity.

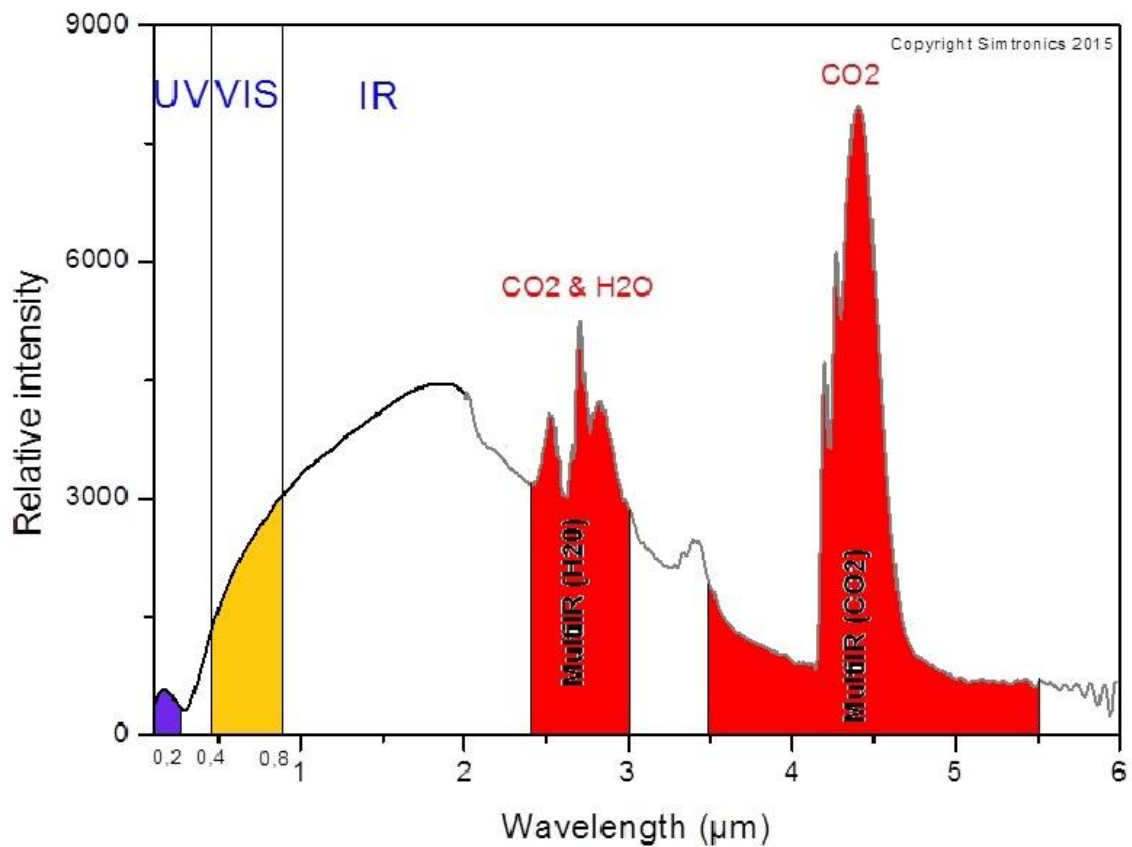
Based on:

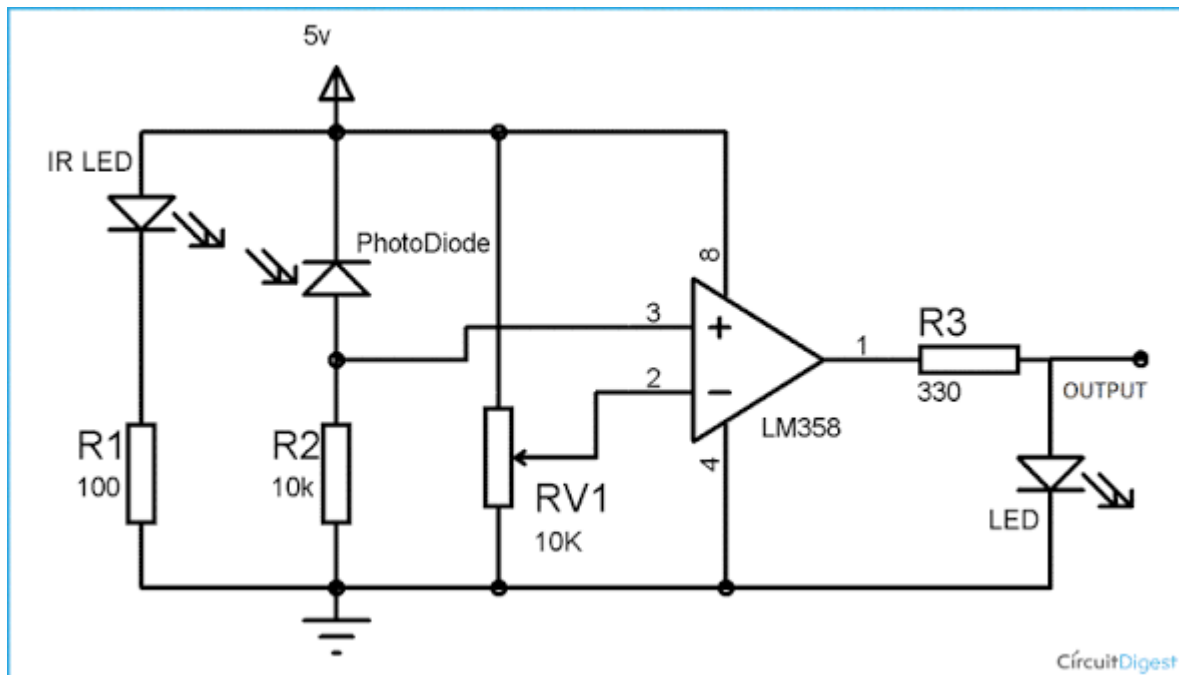
- Surface adsorption
- Redox chemical reactions
- Resistance variation

🔥 🔧 Flame Sensor



Hydrocarbon fire "typical" emission spectrum





4

Scientific Principle:

Infrared Radiation Detection

- Fire emits strong IR radiation (760–1100 nm).
- Sensor uses **IR photodiode / phototransistor**.

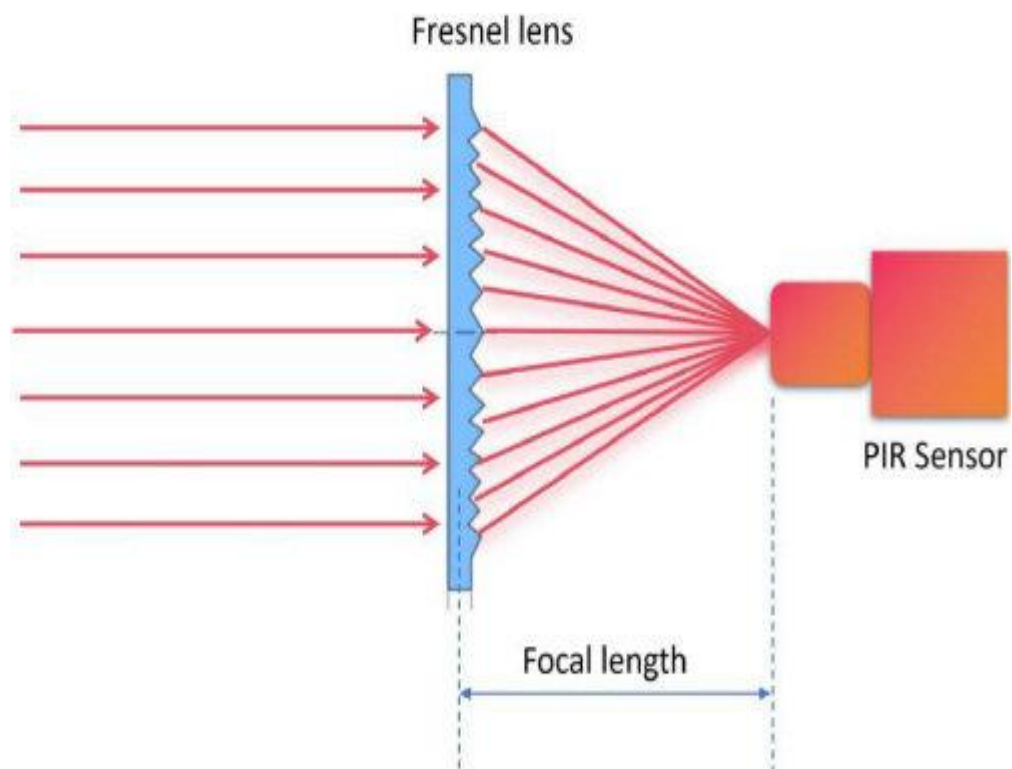
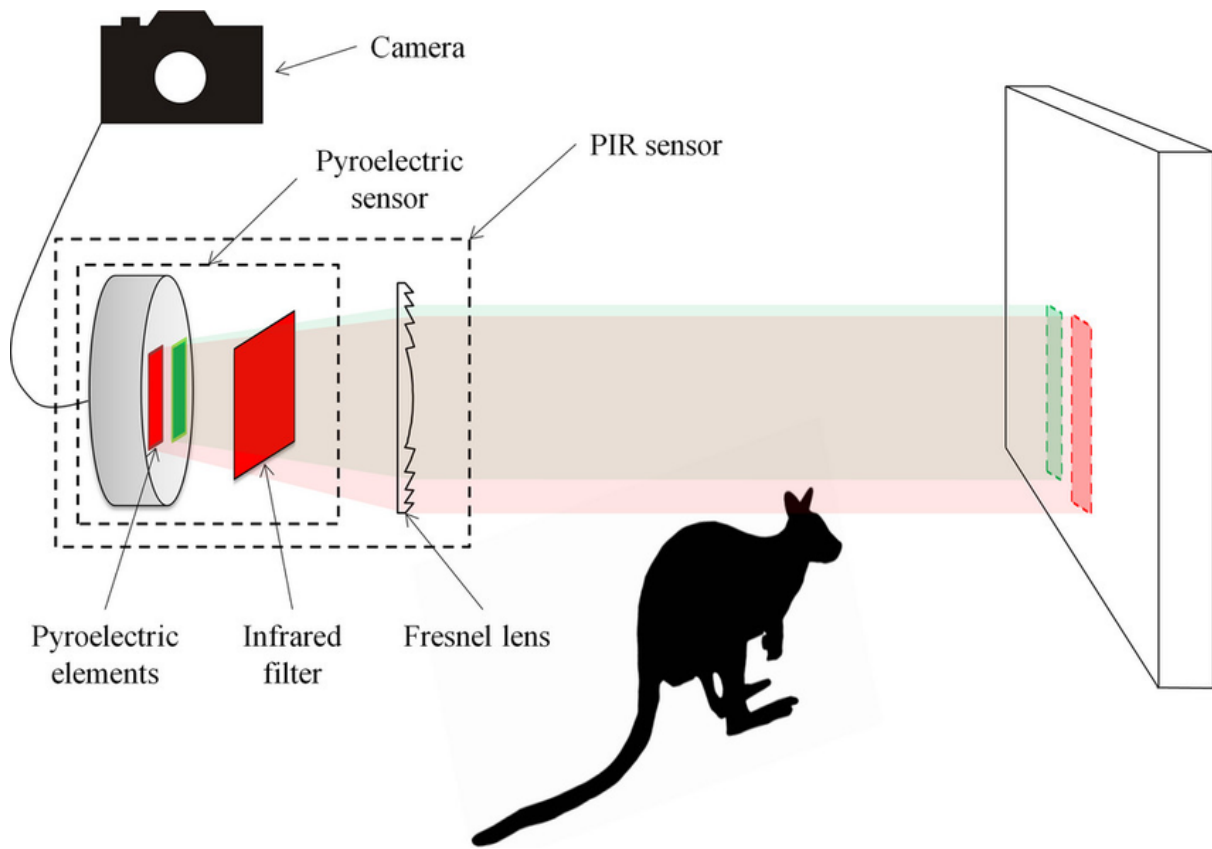
Working:

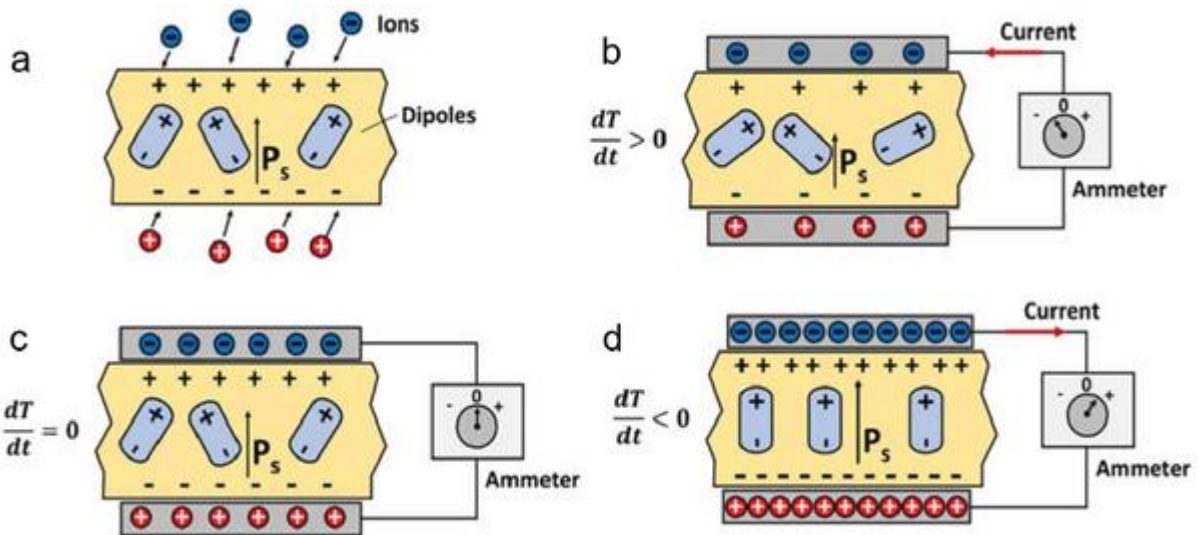
- When IR photons hit sensor:
 - Photoelectric effect occurs.
 - Generates current.
 - Output changes.

Based on:

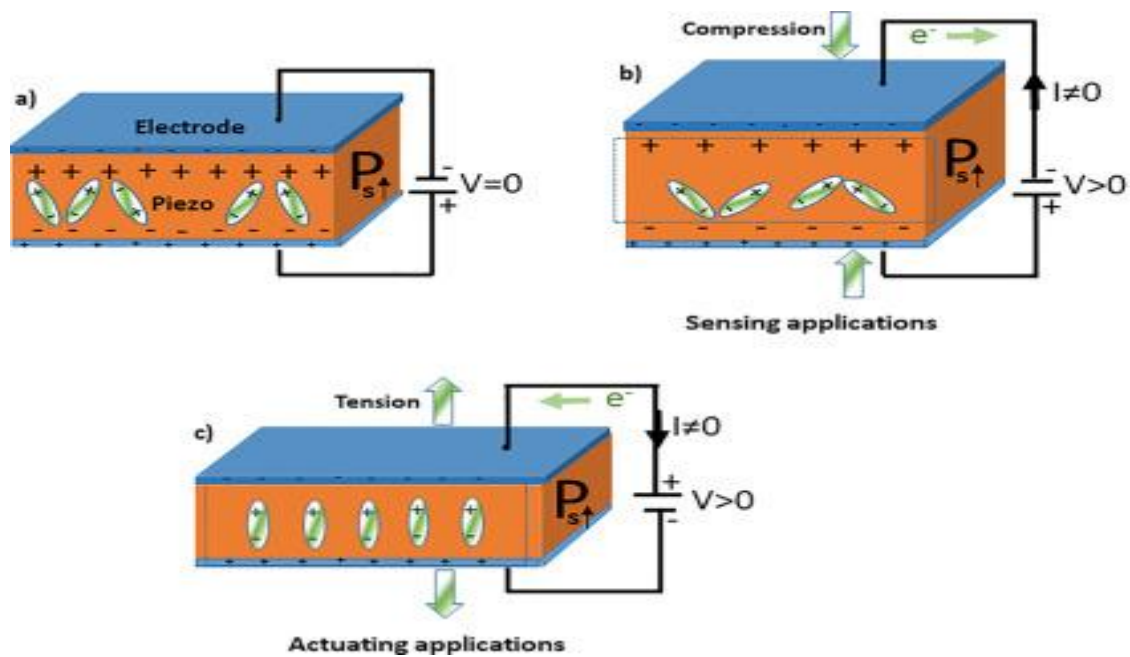
- **Photoelectric effect**
- Electromagnetic radiation absorption

5 PIR Motion Sensor (HC-SR501)





4



 Scientific Principle:

 **Pyroelectric Effect**

- Human body emits IR radiation (~9–10 μm wavelength).
- PIR contains **pyroelectric crystal** (e.g., Lithium Tantalate).

Pyroelectric Effect:

When temperature of crystal changes:

- Electric charge is generated.

📌 Based on:

$$\Delta T \rightarrow \text{Change in Polarization} \rightarrow \text{Voltage Generated}$$

The Fresnel lens focuses IR radiation onto sensor.

Important:

- PIR detects **change in IR**, not constant temperature.
-

🔔 6 Active Buzzer

🔬 Scientific Principle:

✅ Piezoelectric Effect

- Uses piezoelectric crystal.
- When voltage applied:
 - Crystal vibrates.
 - Produces sound waves.

📌 Based on:

Electric field → Mechanical deformation → Sound

💡 7 LED

🔬 Scientific Principle:

✅ Electroluminescence

- Made from semiconductor (P-N junction).
- When forward biased:
 - Electrons recombine with holes.
 - Energy released as photons (light).

📌 Based on:

Electron-hole recombination.

⚡ 8 Capacitor (1000μF)

🔬 Scientific Principle:

✅ Electrostatic Energy Storage

Stores energy in electric field.

$$Q = CV$$

Used to smooth voltage fluctuations.